Kubernetes Deployment Assignment

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Batch 03 - Assignment 3 - Kubernetes

Introduction

This assignment demonstrates deployment and management of an application using Kubernetes. The project includes containerization using Docker, deployment using Kubernetes resources, rolling updates, and health monitoring using probes.

Objective

The main objective of this assignment is to understand:

- Pod creation
- ReplicaSet management
- Deployment management
- Rolling updates
- Kubernetes Services
- Container health checks using probes

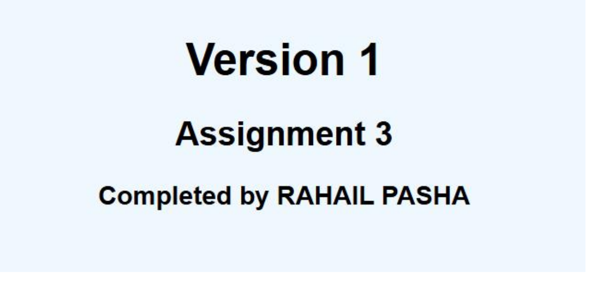
Frontend Application

The application is a simple frontend UI created using HTML and CSS.

Version 1

Displays:

Version 1
Assignment 3
Completed by RAHAIL PASHA

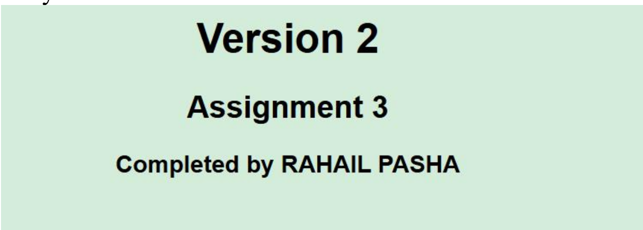


Version 2

Displays:

Version 2

Assignment 3
Completed by RAHAIL PASHA



Docker Setup

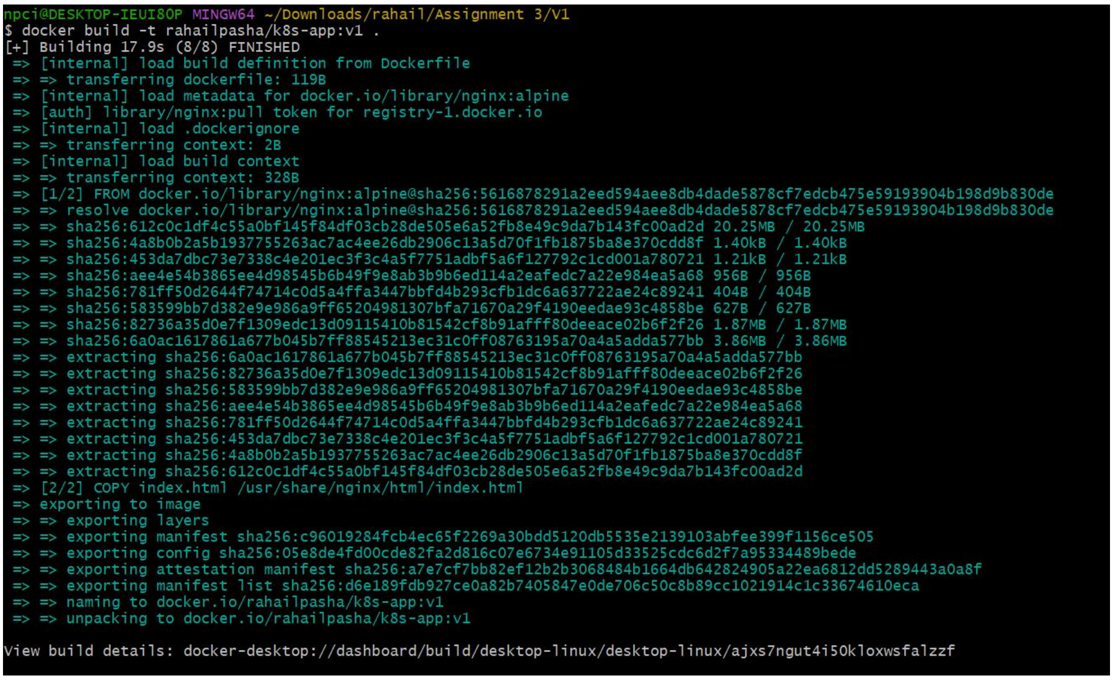
Dockerfile

FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80

Docker Build Commands

Build v1

docker build -t rahailpasha/k8s-app:v1 .



```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/V1
$ docker build -t rahailpasha/k8s-app:v1 .
[+] Building 17.9s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 119B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [auth] library/nginx:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 328B
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
 => => resolve docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
 => => sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 20.25MB / 20.25MB
 => => sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 1.40kB / 1.40kB
 => => sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 1.21kB / 1.21kB
 => => sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68 956B / 956B
 => => sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 404B / 404B
 => => sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be 627B / 627B
 => => sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deeace02b6f2f26 1.87MB / 1.87MB
 => => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB
 => => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb
 => => extracting sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deeace02b6f2f26
 => => extracting sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be
 => => extracting sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68
 => => extracting sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241
 => => extracting sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721
 => => extracting sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f
 => => extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d
 => [2/2] COPY index.html /usr/share/nginx/html/index.html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:c96019284fcb4ec65f2269a30bdd5120db5535e2139103abfee399f1156ce505
 => => exporting config sha256:05e8de4fd00cde82fa2d816c07e6734e91105d33525cdc6d2f7a95334489bede
 => => exporting attestation manifest sha256:a7e7cf7bb82ef12b2b3068484b1664db642824905a22ea6812dd5289443a0a8f
 => => exporting manifest list sha256:d6e189fdb927ce0a82b7405847e0de706c50c8b89cc1021914c1c33674610eca
 => => naming to docker.io/rahailpasha/k8s-app:v1
 => => unpacking to docker.io/rahailpasha/k8s-app:v1

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ajxs7ngut4i50kloxwsfalzzf
```

Build v2

docker build -t rahailpasha/k8s-app:v2 .

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/v2
$ docker build -t rahailpasha/k8s-app:v2 .
[+] Building 1.7s (8/8) FINISHED                                                                 docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                             0.0s
 => => transferring dockerfile: 119B                                                                             0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                  1.4s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                     0.0s
 => [internal] load .dockerignore                                                                                0.0s
 => => transferring context: 2B                                                                                  0.0s
 => [internal] load build context                                                                                0.0s
 => => transferring context: 322B                                                                                0.0s
 => CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de  0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de  0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                       0.0s
 => exporting to image                                                                                           0.1s
 => => exporting layers                                                                                          0.1s
 => => exporting manifest sha256:63c1fe1dd6a7d90541db45887f7b57d9aa856f24a1209512239819dc4c5bde26                0.0s
 => => exporting config sha256:573bed32869d4f92970e4dc3df7c8fc26576e8931a542bc04f2f39944b089444                  0.0s
 => => exporting attestation manifest sha256:ded9699e9debc96c3c1e7affa2a1f6dec024742efc699e07e69aa61fd063a4b3    0.0s
 => => exporting manifest list sha256:41083140b2779613367c4243fb82e2be1b26680d8ba56480d0223f0c0fe93411           0.0s
 => => naming to docker.io/rahailpasha/k8s-app:v2                                                                0.0s
 => => unpacking to docker.io/rahailpasha/k8s-app:v2                                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/hyo0xxtvxd5nzh374e0f32crz
```

Docker Push Commands

docker push rahailpasha/k8s-app:v1

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/v2
$ docker push rahailpasha/k8s-app:v1
The push refers to repository [docker.io/rahailpasha/k8s-app]
781ff50d2644: Pushed
da24845f79ca: Pushed
acb6e3d8d324: Pushed
583599bb7d38: Pushed
453da7dbc73e: Pushed
aee4e54b3865: Pushed
612c0c1df4c5: Pushed
6a0ac1617861: Pushed
82736a35d0e7: Pushed
4a8b0b2a5b19: Pushed
v1: digest: sha256:d6e189fdb927ce0a82b7405847e0de706c50c8b89cc1021914c1c33674610eca size: 856
```

docker push rahailpasha/k8s-app:v2

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/v2
$ docker push rahailpasha/k8s-app:v2
The push refers to repository [docker.io/rahailpasha/k8s-app]
9fe0ce482052: Pushed
5dd5ca9967b0: Pushed
aee4e54b3865: Layer already exists
612c0c1df4c5: Layer already exists
82736a35d0e7: Layer already exists
4a8b0b2a5b19: Layer already exists
781ff50d2644: Layer already exists
453da7dbc73e: Layer already exists
6a0ac1617861: Layer already exists
583599bb7d38: Layer already exists
v2: digest: sha256:41083140b2779613367c4243fb82e2be1b26680d8ba56480d0223f0c0fe93411 size: 856
```

Kubernetes Files

pod.yml

- Creates a single pod
- Runs container using v1 image

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl apply -f pod.yaml
pod/frontend-pod created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl get pods
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0          8s
```

ReplicaSet (rs.yml)

- Maintains multiple pod replicas
- Ensures high availability 0

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl apply -f rs.yaml
replicaset.apps/frontend-rs created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
frontend-rs   3         3         3       6s
```

---

deployment.yml

- Manages application deployment
- Supports rolling updates
- Includes probes

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl apply -f deployment.yaml
deployment.apps/frontend-deployment created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deployment   3/3     3            3           7s

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ k get pods
bash: k: command not found

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ alias k=kubectl

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ k get pods
NAME                                   READY   STATUS    RESTARTS   AGE
frontend-deployment-59654f6584-8fh5w   1/1     Running   0          110s
frontend-deployment-59654f6584-hgp8q   1/1     Running   0          110s
frontend-deployment-59654f6584-znrkj   1/1     Running   0          110s
frontend-pod                           1/1     Running   0          3m48s
frontend-rs-ghptr                      1/1     Running   0          2m3s
frontend-rs-m6hcw                      1/1     Running   0          2m3s
```

---

service.yml

- Exposes application externally
- Routes traffic to pods

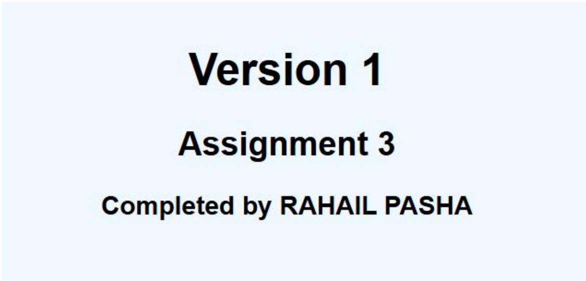
```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ k apply -f service.yaml
service/frontend-service created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ k get svc
NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
frontend-service   NodePort    10.96.39.195   <none>        80:30080/TCP   5s
kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP        3h34m
```

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl port-forward svc/frontend-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

V1 Running

Browser showing:

Version 1
Assignment 3
Completed by RAHAIL PASHA



Update Deployment to v2

Run image update command:

```
kubectl set image deployment/frontend-deployment frontend=rahailpasha/k8s-app:v2
```

This command updates deployment image from:

- v1 → v2

Monitor Rolling Update

Watch update process:

```
kubectl get pods -w
```

During update:

- Old Pods terminate gradually
- New Pods create gradually

Example process:

Old Pod → Terminating
New Pod → Creating
New Pod → Running

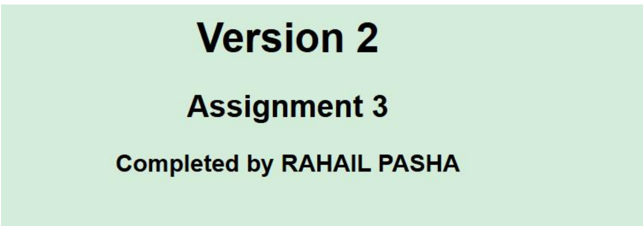
```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ kubectl set image deployment/frontend-deployment frontend=rahailpasha/k8s-app:v2
deployment.apps/frontend-deployment image updated

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/rahail/Assignment 3/k8s
$ k get pods -w
NAME                                   READY   STATUS    RESTARTS   AGE
frontend-deployment-76f4f4cf87-4jtpf   1/1     Running   0          10s
frontend-deployment-76f4f4cf87-pz4pd   1/1     Running   0          10s
frontend-deployment-76f4f4cf87-s5wf9   0/1     Running   0          4s
frontend-pod                           1/1     Running   0          10m
frontend-rs-ghptr                      1/1     Running   0          8m19s
frontend-rs-m6hcw                      1/1     Running   0          8m19s
frontend-deployment-76f4f4cf87-s5wf9   0/1     Running   0          5s
frontend-deployment-76f4f4cf87-s5wf9   0/1     Running   0          5s
frontend-deployment-76f4f4cf87-s5wf9   1/1     Running   0          6s
frontend-rs-ghptr                      1/1     Running   0          8m51s
```

---

v2 Running

Browser showing:

Version 2
Assignment 3
Completed by RAHAIL PASHA



Conclusion

This assignment effectively demonstrated how Kubernetes can be used to deploy, manage, and monitor containerized applications in a production-like environment. The frontend application was successfully containerized with Docker and deployed using Kubernetes components such as Pods, ReplicaSets, Deployments, and Services. Health checks were configured through Liveness, Readiness, and Startup Probes to ensure application stability and reliability.

The deployment process began with Version 1 of the application, which was later upgraded to Version 2 using Kubernetes rolling update functionality. The update was completed smoothly without causing downtime, highlighting Kubernetes' ability to maintain continuous application availability during deployments.

Through this assignment, practical experience was gained in Docker image creation, version management, Kubernetes resource configuration, service exposure, rolling updates, and container health monitoring. Overall, the project provided valuable hands-on understanding of the Kubernetes deployment lifecycle and application management.